KNN with similarity fusion

How to enhance basic knn on collaborative filtering to exploit more available data

Thanina Siouani Léon Hogenmuller Félix de Batz

Setup

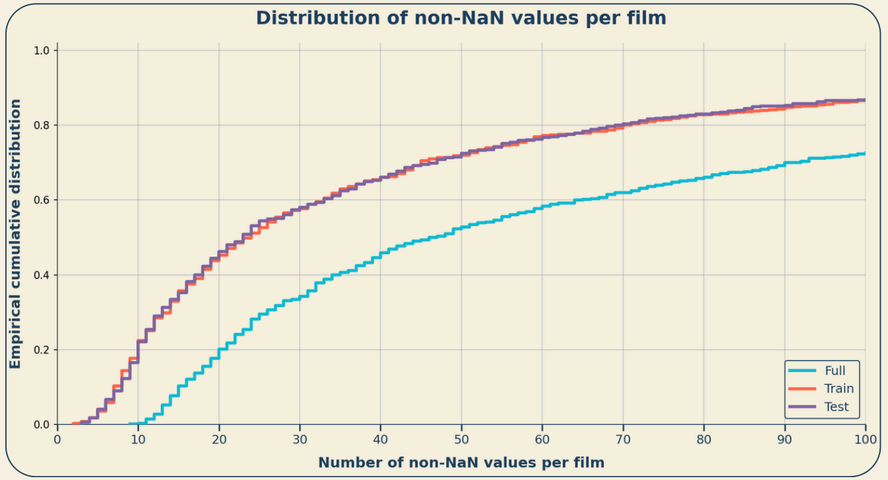
Data sparsity

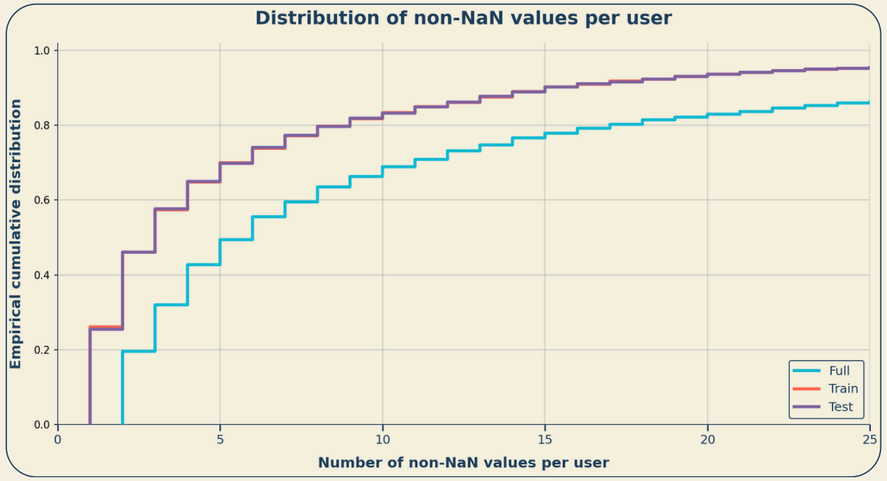
2%

Observed entries

User A	User B	...	User X
	1	...	
...	...	...	4
2		...	

Cold Start

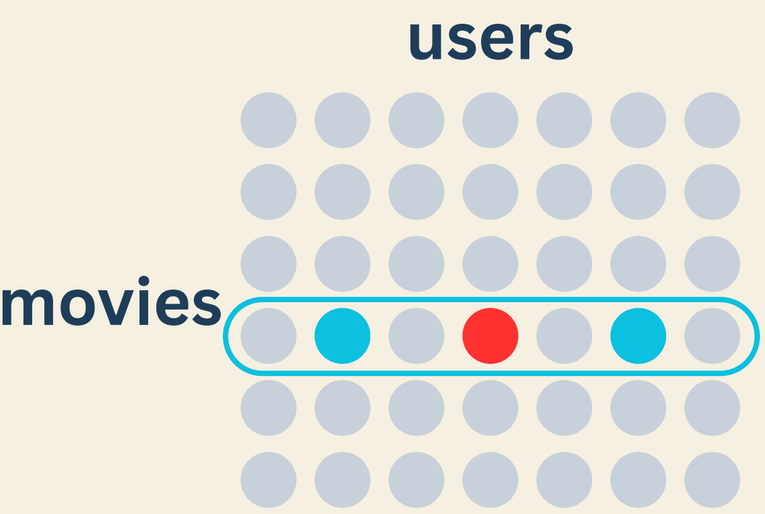




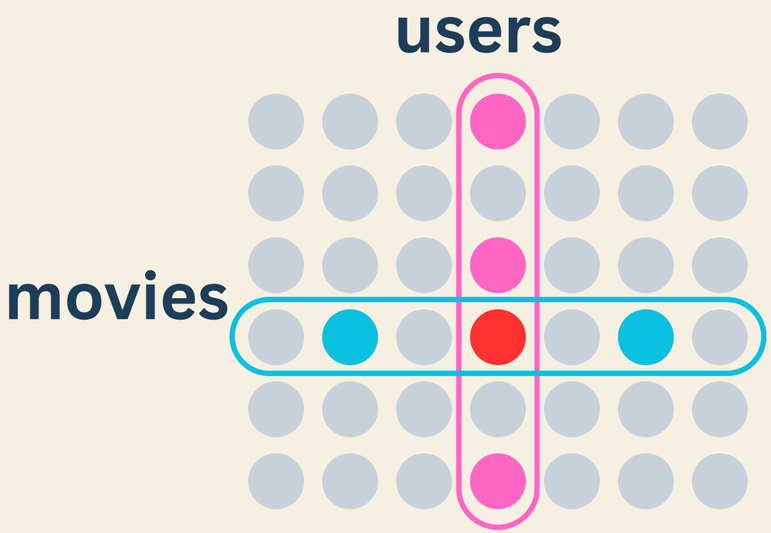
The paper

Unifying User-based and Item-based Collaborative Filtering Approaches by Similarity Fusion

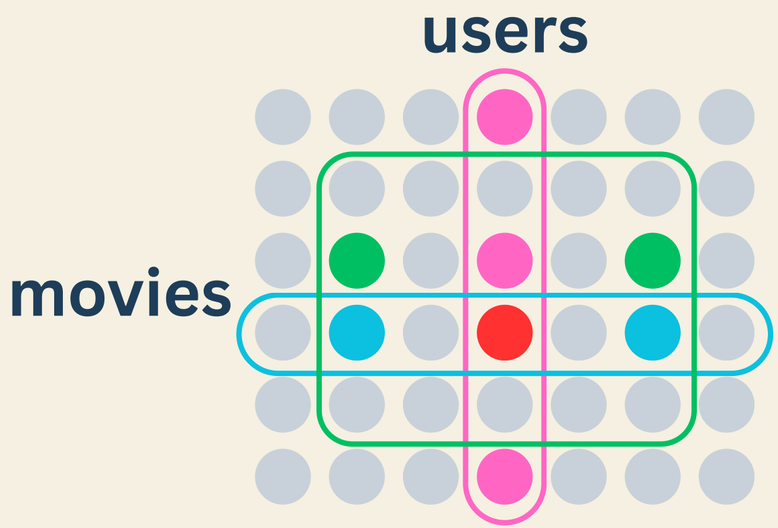
(Jun Wan, Arjen P. de Vries, Marcel J.T. Reinders)



user Knn



user Knn + movie Knn



user Knn + movie Knn + fusion

The paper

Similarity measure

$$s_{ui} = \frac{1}{\sqrt{\left(\frac{1}{s_u}\right)^2 + \left(\frac{1}{s_i}\right)^2}}$$

- s_i : the similarity between two items.
- s_u : the similarity between two users.
- s_{ui} : the similarity user-item

Our take

Aspect	Paper's	Ours
Number of Neighbors	N constant in ratings	k_movie, k_user constant in neighbors
Similarity computation	Cosine similarity on ratings	Cosinus similarity on the deviation
Movies' features	Not available	Similarity based on release year and genres

Our take

Similarity computation

User A	User B
1	5
3	3
5	1

Compute the deviation

User A	User B
-2	+2
0	0
+2	-2

0.23 similarity

-0.5 similarity

Movies' features

$$S_f = w \times S_g + (1-w)\, S_y$$

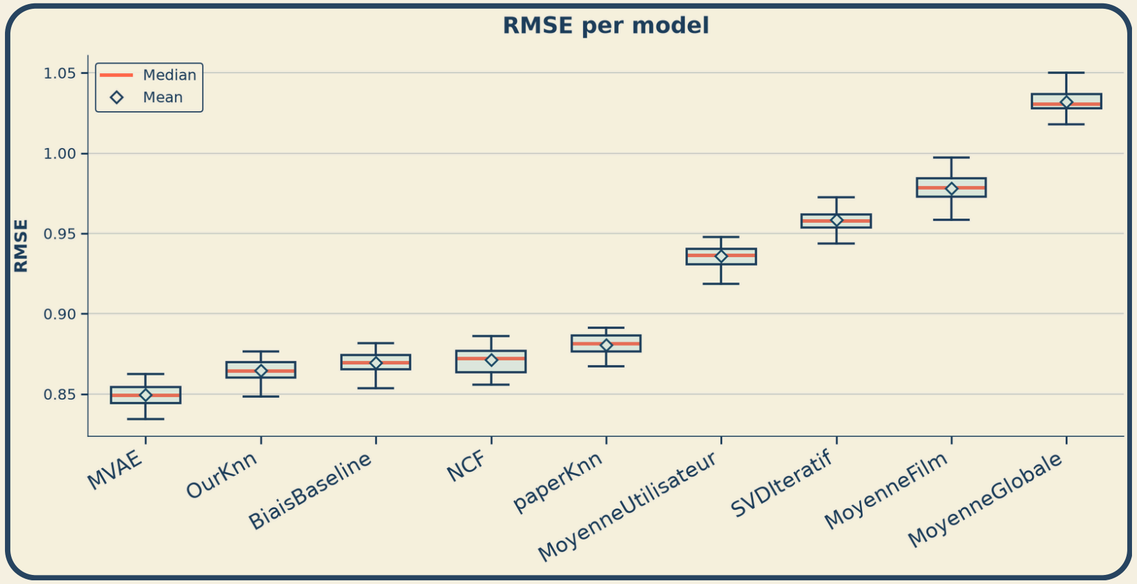
$$S_g(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

$$S_y(a,b) = e^{\frac{|a-b|}{\tau}}$$

$$S_m = \alpha \times S_r + (1-\alpha)\, S_f$$

Results

Score comparison

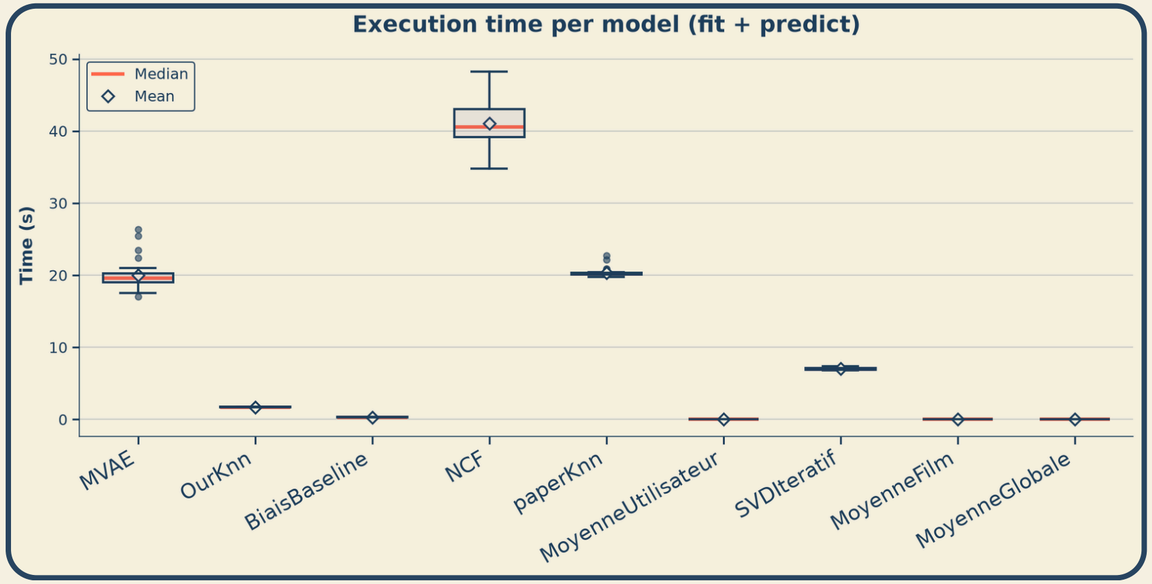


RMSE Mean SVD Iteratif = 0.9584

RMSE Mean Knn = 0.8645

RMSE Mean MVAE = 0.8497

Time comparison



Time Mean SVD Iteratif = 7.014 s

Time Mean Knn = 1.714 s

Time Mean MVAE = 19.966 s

Conclusion

Limits of the KNN with similarity fusion

What’s next?

MVAE improvements



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